Mark A. Kheyfets

Worcester 01609
(617)-669-1221 | mkheyfets@wpi.edu
https://mkheyfets-tech.github.io/robotics-portfolio/

EDUCATION

Worcester Polytechnic Institute ***Worcester, MA 01609***
B.S., Robotics Engineering Expected Graduation: ***May 2027***
M.S., Robotics Engineering Expected Graduation: ***May 2028***
GPA: 3.6

SKILLS

- **Programming Languages**: Python, Racket, MATLAB, C, C++
- **Robotics and Engineering:** Inverse kinematics, Trajectory Planning, Image Processing, FEM Analysis
- **Software:** GitHub, SolidWorks, Fusion 360, Streamlit, MS Office
- **Hardware:** Cameras, IR Sensors, Ultrasonic Sensors

PROFESSIONAL EXPERIENCE

Research Analyst, OpenStar Technologies ***Spring 2026***

- Conducted mixed-methods research on public sentiment toward nuclear fusion in New Zealand's energy sector.
- Performed data analysis and visualization using Streamlit, identifying key trends in fusion familiarity and public support.
- Contributed to development of an evidence-based outreach policy focused on public engagement, audience-first storytelling, and legislative outreach.

Resident Advisor, Worcester Polytechnic Institute ***Fall 2025-Spring 2026***

- Served as a peer leader and primary resource for students in residential housing, fostering an inclusive and supportive community.
- Managed community-building initiatives, campus policy enforcement, crisis response, and residential programming to enhance student well-being.

PROJECTS

Vision Guided Robotic Pick-and-Place System:

Real-Time Object Detection and Autonomous Sorting using Computer Vision and Kinematics.

- Built an autonomous robotic sorting system using computer vision, forward and inverse kinematics, and quintic trajectory planning.
- Implemented masking, color thresholding, centroid detection, fisheye correction, and image-to-robot coordinate transformations for reliable object localization.
- Troubleshot calibration, sensing, and motion-planning issues to improve execution accuracy and repeatability.

Romi Autonomous Navigation and AprilTag Detection System:

Differential-Drive Mobile Robot with Forward/Inverse Kinematics and Computer Vision.

- Implemented second-order forward kinematic pose estimation, trajectory control, and autonomous path planning for accurate real-time localization and navigation.
- Integrated OpenMV computer vision and AprilTag detection for autonomous target searching, alignment, and docking using proportional control and state-machine-based behaviors.
- Developed proportional control routines for tag alignment and autonomous docking, including searching and recovery behaviors when targets were lost.

Translational Parallel Manipulator:

Autonomous Warehouse Robot with Parallel Manipulator.

- Designed and programmed a 2-DOF translational parallel manipulator for autonomous box pick-and-place on multi-level warehouse shelves, integrating one brushed DC motor and two servo motors with inverse kinematics to maintain constant gripper orientation.
- Integrated one brushed DC motor and two servo motors using inverse kinematics to maintain constant parallel gripper orientation.
- Developed custom C++ firmware for Romi base navigation, manipulator motion, and gripper actuation via serial-based control.